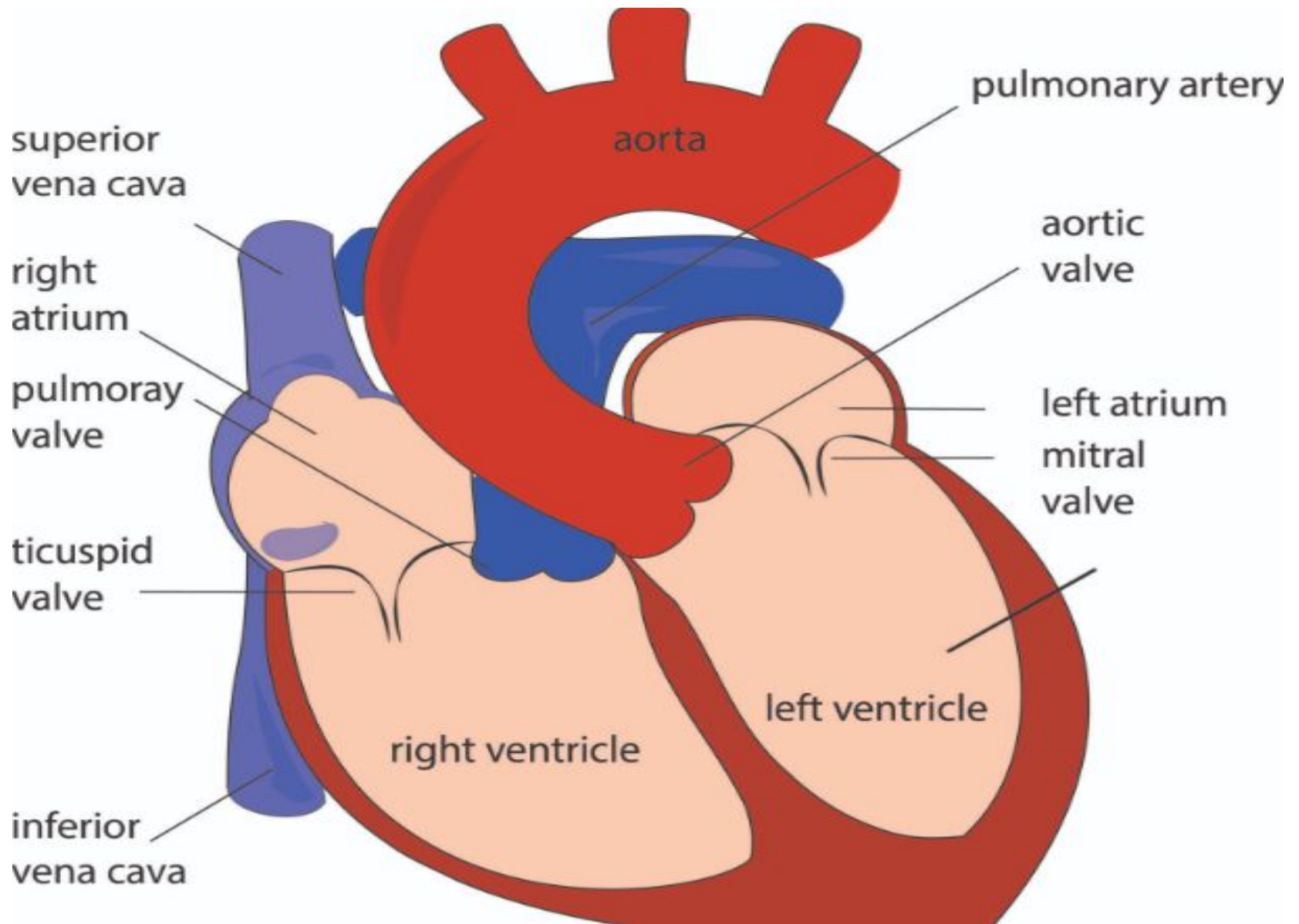


STRUCTURE OF HEART



The heart is a vital, fist-sized muscular organ located slightly on the left side of the chest. It consists of four main chambers: two atria and two ventricles.

The heart is a crucial organ that functions as the body's pump, ensuring blood circulation throughout the body. It consists of four main chambers

Left and right atria (upper chambers)

Left and right ventricles (lower chambers)

These chambers work in a coordinated manner to receive oxygen-poor blood, pump it to the lungs for oxygenation (adding blood to oxygen), and then distribute oxygen-rich blood to the rest of the body. The heart's structure also includes valves that prevent backflow and ensure blood flows in the correct direction.

The heart has three layers of tissue:

Endocardium: The innermost layer, provides a smooth lining for chambers and valves

Myocardium: The middle layer, composed of muscle tissue that enables heart contractions

Epicardium: The outermost layer protects the heart and reduces friction with surrounding structures.

Pericardium

The [pericardium](#) is a double-walled sac that encloses the heart.

Coronary Arteries and Veins

[Coronary arteries](#) are blood vessels that supply the heart muscle (myocardium) with oxygen-rich blood. Coronary veins remove oxygen-poor blood.

Major Blood Vessels:

Major blood vessels of the heart include:

[Aorta](#): The largest artery in the body, carrying oxygen-rich blood from the left ventricle to the body.

[Pulmonary arteries](#): Vessels that carry oxygen-poor blood from the right ventricle to the lungs.

Pulmonary veins: Vessels that carry oxygen-rich blood from the lungs to the left atrium.

[Superior vena cava](#) and [inferior vena cava](#): Carry oxygen-poor blood from the body to the right atrium

The internal structure of the heart is designed to facilitate its function as a powerful pump.

The heart has four chambers, including

Right atrium: Receives oxygen-poor blood from the body through the superior and inferior vena cava.

Right ventricle: Pumps the oxygen-poor blood to the lungs via the pulmonary artery.

Left atrium: Receives oxygen-rich blood from the lungs through the pulmonary veins.

Left ventricle: Pumps the oxygen-rich blood to the rest of the body through the aorta.

The heart's valves that prevent backflow and ensure that the blood continues to flow in the right direction include:

Tricuspid valve: Located between the right atrium and right ventricle, the tricuspid valve has three flaps (cusps) that open to allow blood to flow from the right atrium to the right ventricle and close to prevent blood from flowing backward.

Pulmonary valve: Positioned between the right ventricle and the pulmonary artery, this valve opens to let blood flow from the right ventricle into the pulmonary artery, which leads to the lungs. It closes to prevent blood from returning to the right ventricle.

Mitral valve: The mitral valve has two flaps (cusps) between the left atrium and left ventricle. It opens to allow oxygen-rich blood from the left atrium to flow into the left ventricle and closes to prevent backflow into the atrium.

Aortic valve: Found between the left ventricle and the aorta, the aortic valve opens to allow blood to flow from the left ventricle into the aorta. This main artery carries oxygen-rich blood to the rest of the body. It closes to prevent blood from flowing back into the left ventricle.